

Nicholas Diamond

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EDUCATION

The University of Tennessee, Haslam College of Business

Bachelor of Science in Business Administration

Major: *Business Analytics* - Collateral: *Economics*

GPA: 3.59 (Major) / 3.16 (Overall)

Knoxville, TN

Graduation: December 2025

PROJECT EXPERIENCE

Hidden Gem Movie Recommender

R · Shiny · arules · Market Basket Analysis

Live app: ndiamond7.shinyapps.io/Movie-Recommendation-App/

- Built a market basket analysis model on large-scale MovieLens ratings data to generate personalized film recommendations, deployed as a live interactive Shiny web application.
- Designed and customized the full UI/UX; authored a complete user manual documenting app functionality, filter controls, and result interpretation.

Diamond Pricing Analysis - BrightStone Jewelers

R · randomForest · DALEX · Executive Communication

- Engaged as analytics consultant to diagnose mispricing across 59,000 diamonds; built and tuned a random forest achieving 97.4% accuracy and reducing RMSE by 84% over the naive baseline.
- Applied Shapley value analysis to translate model outputs into explainable attribute-level pricing contributions; delivered client-facing presentation with procurement, marketing, and dynamic pricing recommendations.

Return Rate Analysis - ShopLC

R · randomForest · Executive Communication

- Cleaned and modeled 617,904 weekly sales records across 40 product families to establish expected return rate benchmarks, reducing prediction error by 35% over the naive baseline.
- Identified product families deviating most from model expectations and delivered an executive-ready brief with prioritized quality and operations recommendations.

YouTube Ad Monetization Strategy

Tableau · Engagement Analysis · Regional Strategy

- Analyzed official YouTube platform data across viewership, engagement, and regional dimensions to identify ad monetization opportunities by content category and market.
- Built a Tableau visualization suite and delivered a strategic ad placement framework distinguishing high-volume from high-engagement categories to a consulting-style audience.

Business Data Pipelines - Alteryx

Alteryx · SQL Logic · Relational Databases

- Designed multi-table Alteryx workflows serving HR, Marketing, Supply Chain, and Sales departments, including bonus calculations, customer segmentation, reorder alerts, and top-performer rankings.

Paper Copter Experiment - Full Factorial DOE

JMP · Executive Communication · Regression

- Validated measurement consistency across groups using S-charts, built a regression model in JMP to identify design factors driving flight time, and delivered a written executive summary with an optimal design recommendation.

APIs & LIVE DATA INTEGRATIONS

Portfolio site (diamondnp9.github.io) includes three live REST API integrations with server-side caching via GitHub Actions, encrypted secret management, and dynamic DOM rendering with JavaScript.

- **World Bank Explorer** - Interactive choropleth map pulling live economic and social indicators by country from the World Bank public API.
- **Live Market Snapshot** - Real-time equity data via Finnhub API, refreshed every 15 minutes on weekdays using a scheduled GitHub Actions workflow.
- **NASA APOD Integration** - Daily astronomy content served from NASA's public APOD API with dynamic DOM rendering.

WORK EXPERIENCE

The House, Inc.

Operations Intern

Woodbridge, VA

July 2025 – August 2025

- Assisted in managing a team of 15 volunteers and employees by assigning tasks, tracking progress, and supporting daily operations.
- Developed financial goals and targets by analyzing donation trends, the financial well-being of target organizations, and income from state and local grants.
- Optimized workflow efficiency through real-time problem-solving and clear cross-functional communication.

Lowe's Companies, Inc.

Retail Sales Associate, Garden & Outdoor

Knoxville, TN

April 2024 – November 2024

- Assisted customers in the garden and outdoor section by providing product knowledge and recommendations to support purchase decisions.
- Operated a forklift to transport and stage inventory, maintaining an organized and hazard-free outdoor department.
- Upheld safety protocols and store standards in a high-traffic retail environment with zero workplace incidents.

Firebirds

Server

Gainesville, Virginia

May 2023 – January 2024

- Delivered high-quality dining experiences by efficiently managing 4-10 tables, ensuring timely and accurate service.
- Suggested menu items based on guest preferences, increasing upselling opportunities and overall customer satisfaction.

Landini Brothers

Server

Alexandria, Virginia

June 2022 – August 2022

- Provided exceptional service in an upscale dining setting by attentively managing guest needs and special requests.
- Demonstrated expert menu knowledge to recommend food and wine pairings, increasing customer satisfaction.

Rockwood Bar and Grill

Busser & Server

Gainesville, VA

August 2020 – June 2021

- Promoted to server after demonstrating strong knowledge of the menu and excellent customer service skills.
- Managed a high-volume section, ensuring timely service and enhancing the guest experience.

TECHNICAL SKILLS

- **Tableau** - Built dashboards to visualize key metrics and business trends; applied in multiple academic projects focused on reporting and communication with non-technical audiences.
- **Alteryx** - Streamlined data preparation and automation tasks to support accurate, efficient analysis and reporting
- **R & RStudio** - Used for statistical analysis, trend identification, and pattern recognition in support of data-driven business decisions
- **JMP** - Applied in experimental design and data analysis for course projects; used to validate insights and support quality decision-making
- **Excel & Office 365** - Proficient in organizing, cleaning, and analyzing large datasets; skilled in using formulas, pivot tables, and visualizations to support cross-functional reporting
- **Power BI** - Introductory knowledge with ongoing learning; familiar with dashboard concepts and integration with Excel and business data
- **Stata** - Utilized in academic projects for regression and data manipulation, supporting analytical research and reporting
- **HTML** - Basic understanding used to format and structure web-based content for reporting and presentation purposes
- **SQL** - Used to query, clean, and join large datasets across multiple tables; applied in academic projects to build relational databases, perform exploratory analysis, and support data-driven decision-making with efficient data extraction

LEADERSHIP EXPERIENCE

United States Naval Academy Leadership Program

Attendee

Annapolis, Maryland

June 2018

- Developed leadership and teamwork skills through intensive training exercises and workshops.
- Applied strategic decision-making in high-pressure scenarios, strengthening problem-solving abilities